Q1.

(a) $\frac{1}{x}$ = cos$y$ or $\frac{1}{y}$ = cos$x$

M1

$y = \cos^{-1}\frac{1}{x}$ ie result

CSO

A1

(b) $\frac{\mathrm{d}}{\mathrm{d}x}\left(\sec^{-1}x\right)=\frac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\frac{1}{x}\right)$

M1

$= -\frac{1}{\sqrt{1-\frac{1}{x^2}}}$ if in terms of u A0

A1

$\times\left(-\frac{1}{x^2}\right)$

A1

$= \frac{1}{\sqrt{x^4-x^2}}$

clearly shown (AG)

A1

Alternative:

cos $y = \frac{1}{x}$

Use of sec y = x M0

$-\sin y\ \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-1}{x^2}$

(M1)
(A1)

Substitute for sin y

(A1)

Result

(A1)

4

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Q2.

(a) $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}\left(1 - x^2\right)^{-\frac{1}{2}}$

B1

$\times (-2x)$

B1

2

(b) $\int \sin^{-1} x \mathrm{d}x = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} \mathrm{d}x$

A1 for each part of the integration by parts

M1

A1A1

$\int -\frac{x}{\sqrt{1-x^2}} \mathrm{d}x = \sqrt{1-x^2}$ used

ft sign error in $\frac{\mathrm{d}u}{\mathrm{d}x}$

A1F

$\frac{\sqrt{3}}{2}\frac{\pi}{3} + \sqrt{1-\frac{3}{4}} - 1$

substitution of limits

m1

$\frac{1}{6}\sqrt{3}\pi - \frac{1}{2}$

CAO

A1

6

[8]

Q3.

(a) $\frac{x}{1+x^2} + \tan^{-1} x$

B1B1

2

(b) $\int_0^1 \tan^{-1} x \, dx = \left[x \tan^{-1} x\right]_0^1 - \int_0^1 \frac{x \, dx}{1+x^2}$

either use of part (a) or integration by parts. Allow if sign error

M1

$\int \frac{x \, dx}{1+x^2} = \frac{1}{2} \ln(1+x^2)$

ft on $\int \frac{x}{1-x^2} dx$

M1A1F

I = 1 tan−1 $1 - \frac{1}{2} \ln 2$

M1

$= \frac{\pi}{4} - \ln\sqrt{2}$

AG

A1

5

[7]

Examiner reports

Q1.

There were not many convincing solutions to part (a) of this question as there was clearly much confusion between the inverse of a function and the reciprocal of a function. There was also, surprisingly, few complete solutions to part (b). The chain rule for differentiation was simply not applied and many students, after writing $\dfrac{-1}{\sqrt{1-\dfrac{1}{x^2}}}$ expression to the printed answer.

Q2.

The differentiation in part (a) was well done, apart from the occasional sign error. In part (b), whilst most candidates were familiar with integration by parts, many candidates stalled at the second integral, even though the hint had been clearly given in part (a). Even when candidates knew how to finish the question, errors of sign spoilt what would otherwise have been a good solution.

Q3.

This is the first time that a question has been set on inverse trigonometrical functions since this topic was included in the MFP2 specification. It was clear that many candidates did not know what $\tan^{-1}x$ was. They were able to complete part (a) with the help of the formulae booklet although even then there was confusion between the derivatives of $\tan^{-1}$ and $\tanh^{-1} x$ as the derivative of $\tan^{-1}$ was given as $\dfrac{1}{1-x^2}$.

However it was part (b) that revealed the true lack of understanding of inverse trigonometrical functions. Part (b) was either abandoned altogether or when attempted $\tan^{-1} x$ was frequently written as $\dfrac{1}{\tan x}$.